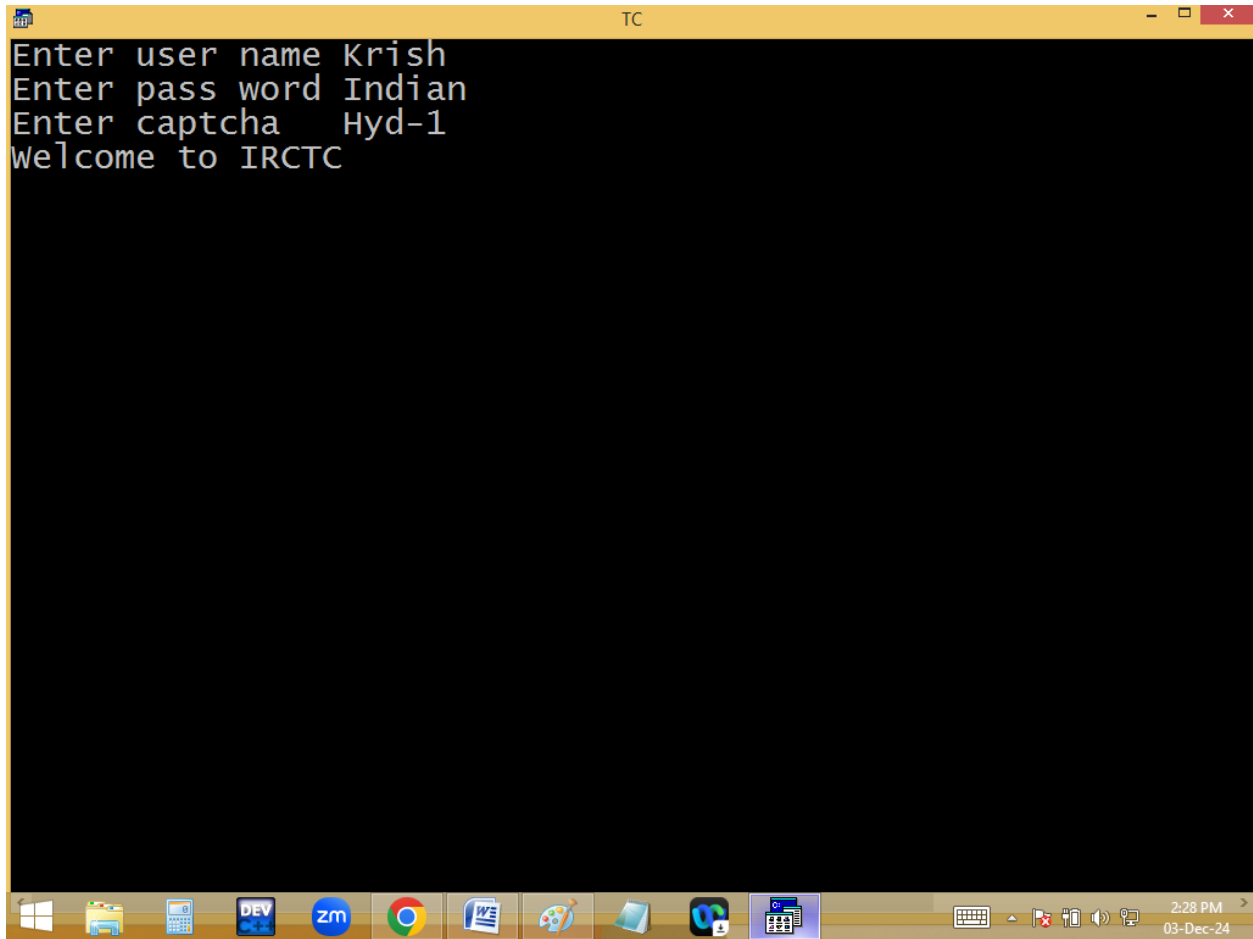


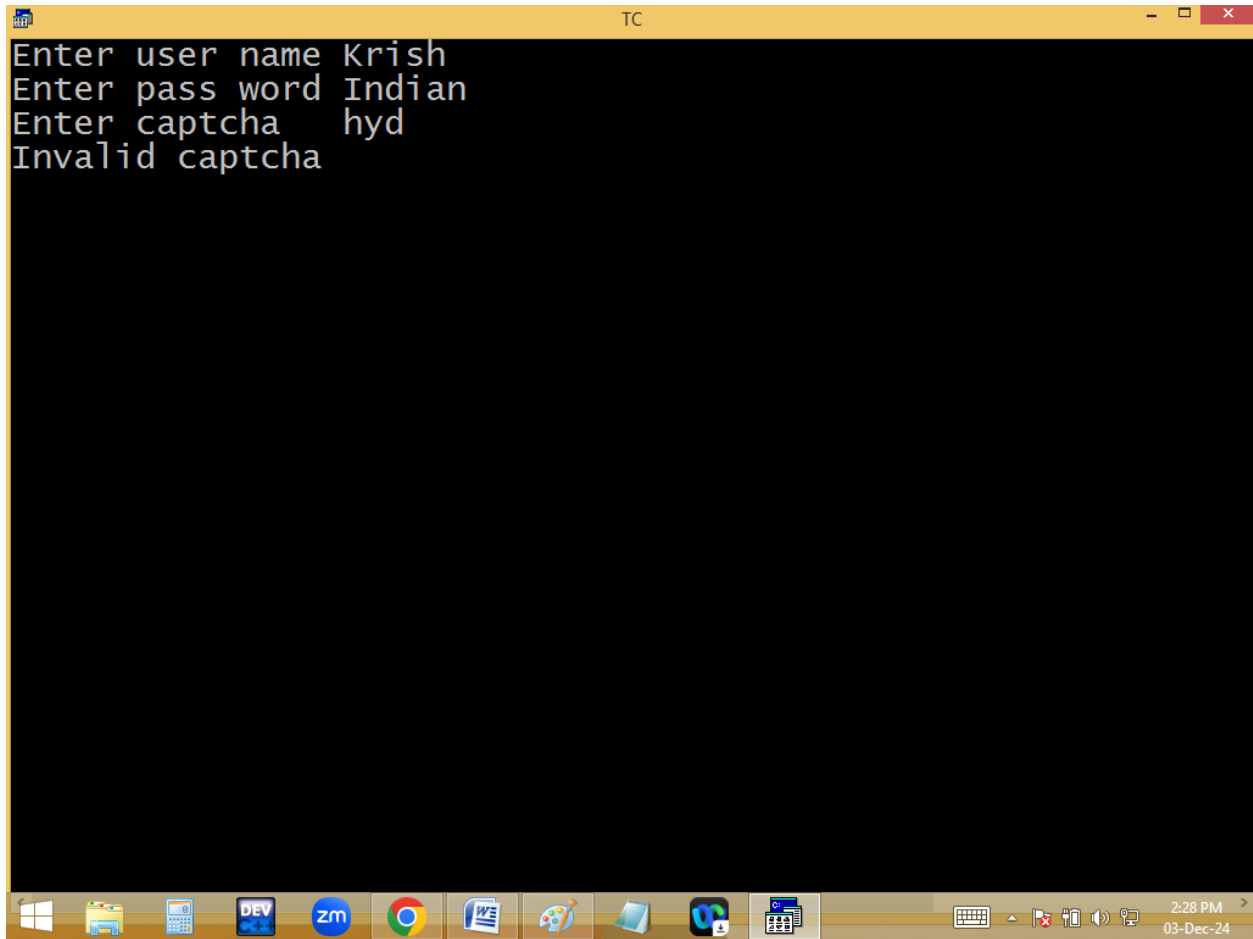
IRCTC Login:

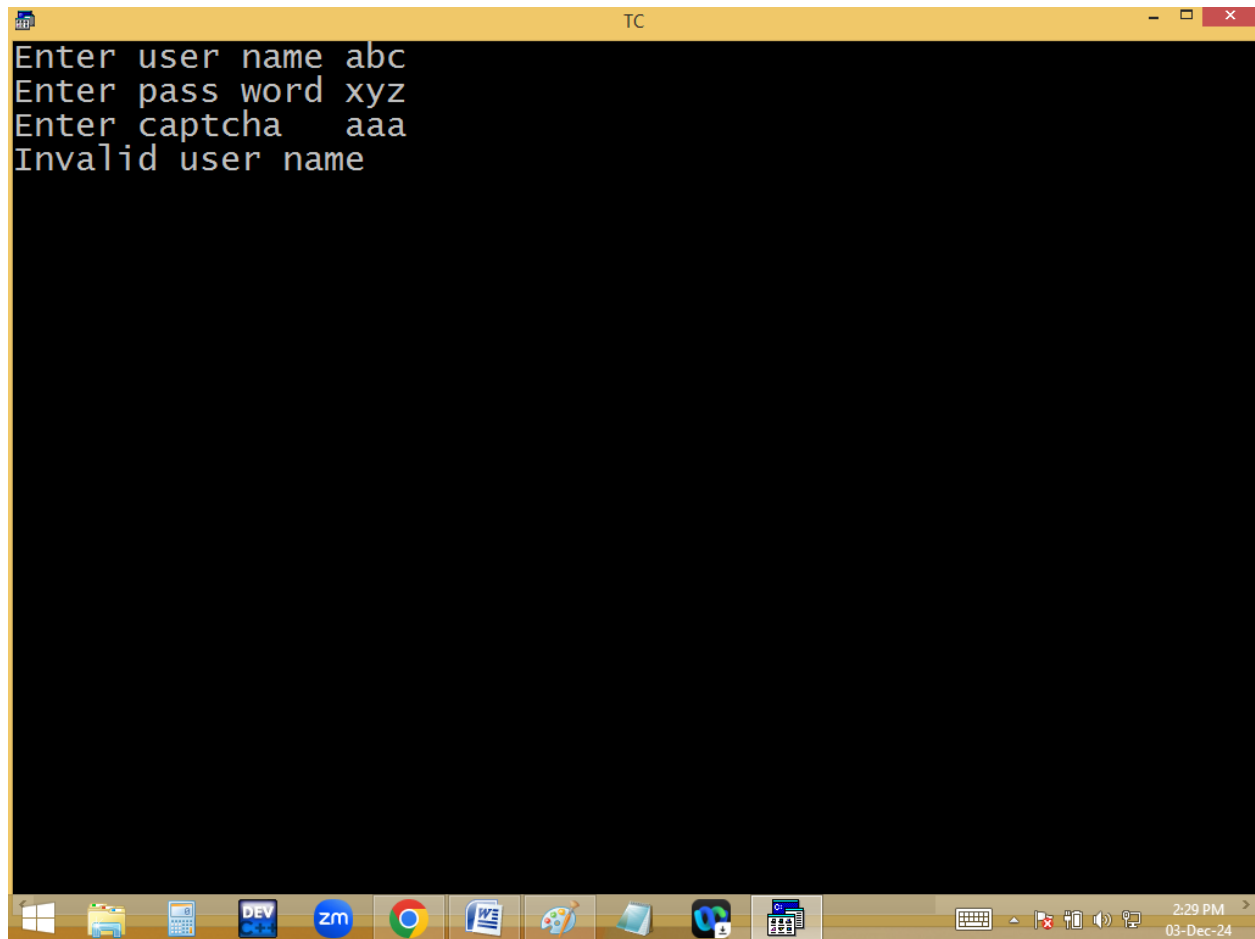
```
TC
File Edit Run Compile Project Options Debug
Line 21 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char user[20], pass[20], cap[20];
clrscr();
printf("Enter user name "); scanf("%s",user);
printf("Enter pass word "); scanf("%s",pass);
printf("Enter captcha "); scanf("%s",cap);
if(strcmp(user,"Krish")==0)
{
if(strcmp(pass,"Indian")==0)
{
if(strcmp(cap,"Hyd-1")==0)puts("welcome to IRCTC");
else puts("Invalid captcha");
}
else puts("Invalid pass word");
}
else puts("Invalid user name");
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

2:28 PM 03-Dec-24





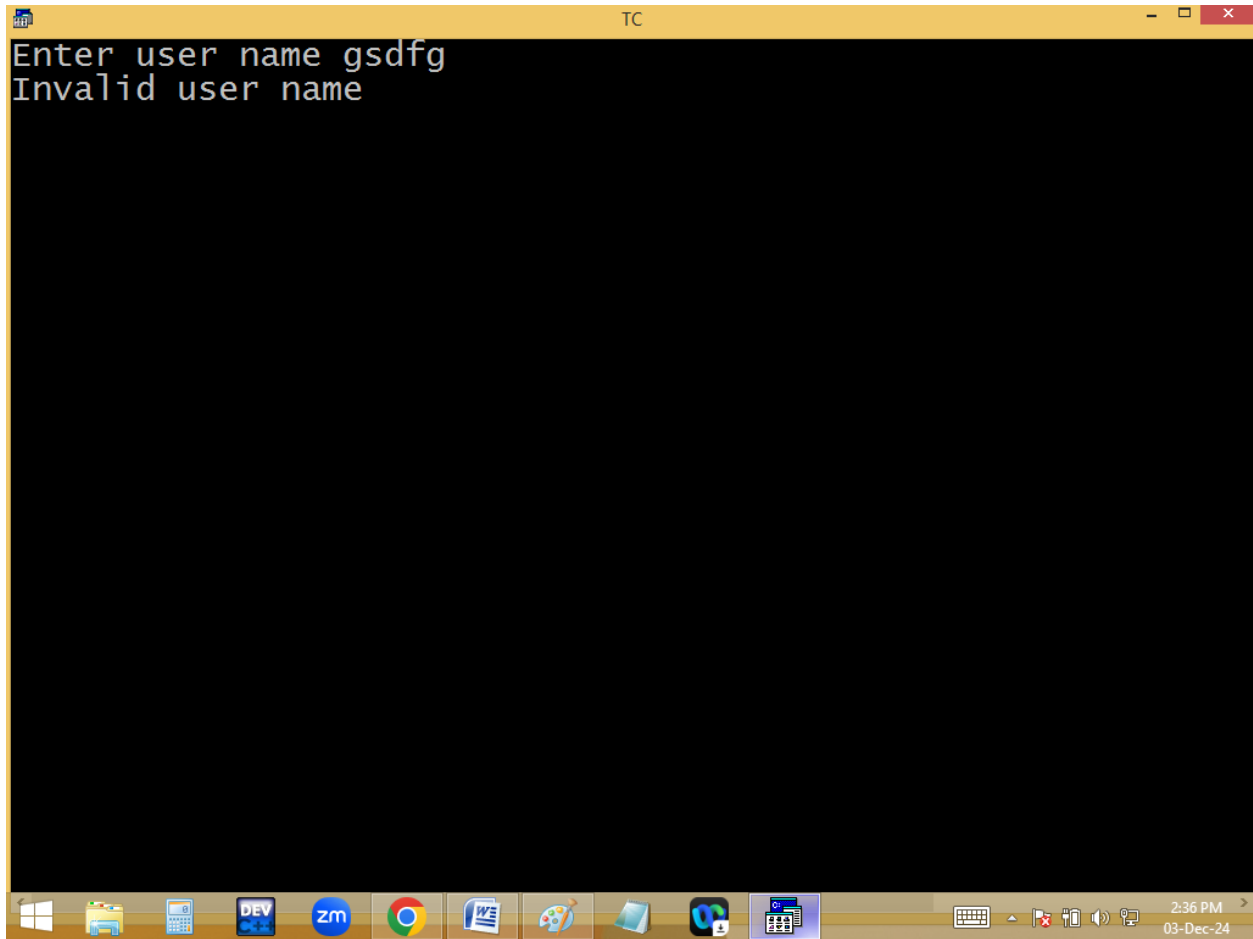


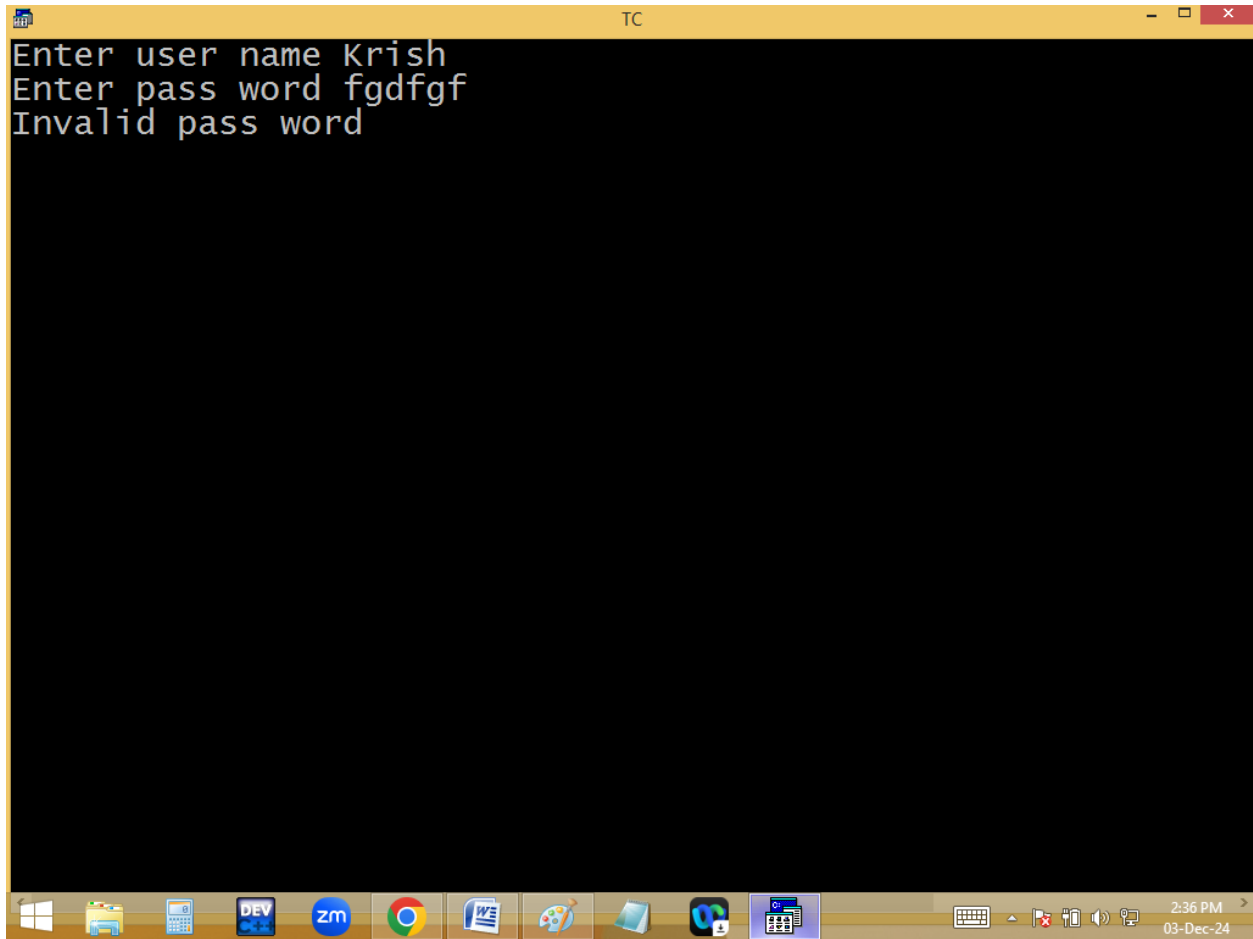
Gmail style login:

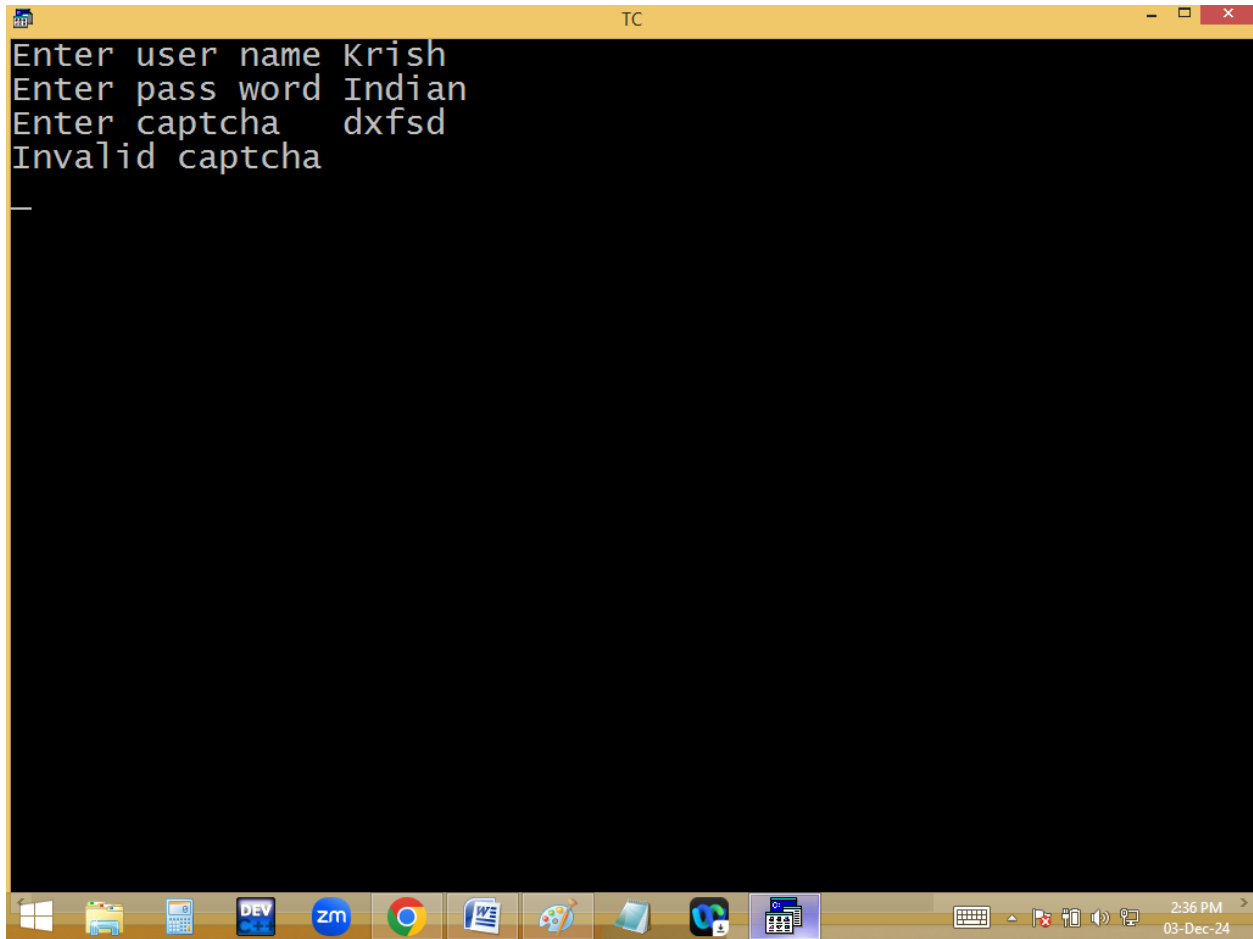
```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 49 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char user[20], pass[20], cap[20];
clrscr();
printf("Enter user name "); scanf("%s",user);
if(strcmp(user,"Krish")==0)
{
printf("Enter pass word "); scanf("%s",pass);
if(strcmp(pass,"Indian")==0)
{
printf("Enter captcha "); scanf("%s",cap);
if(strcmp(cap,"Hyd-1")==0)puts("Welcome to Gmail");
else puts("Invalid captcha");
}
else puts("Invalid pass word");
}
else puts("Invalid user name");
getch();
}
```

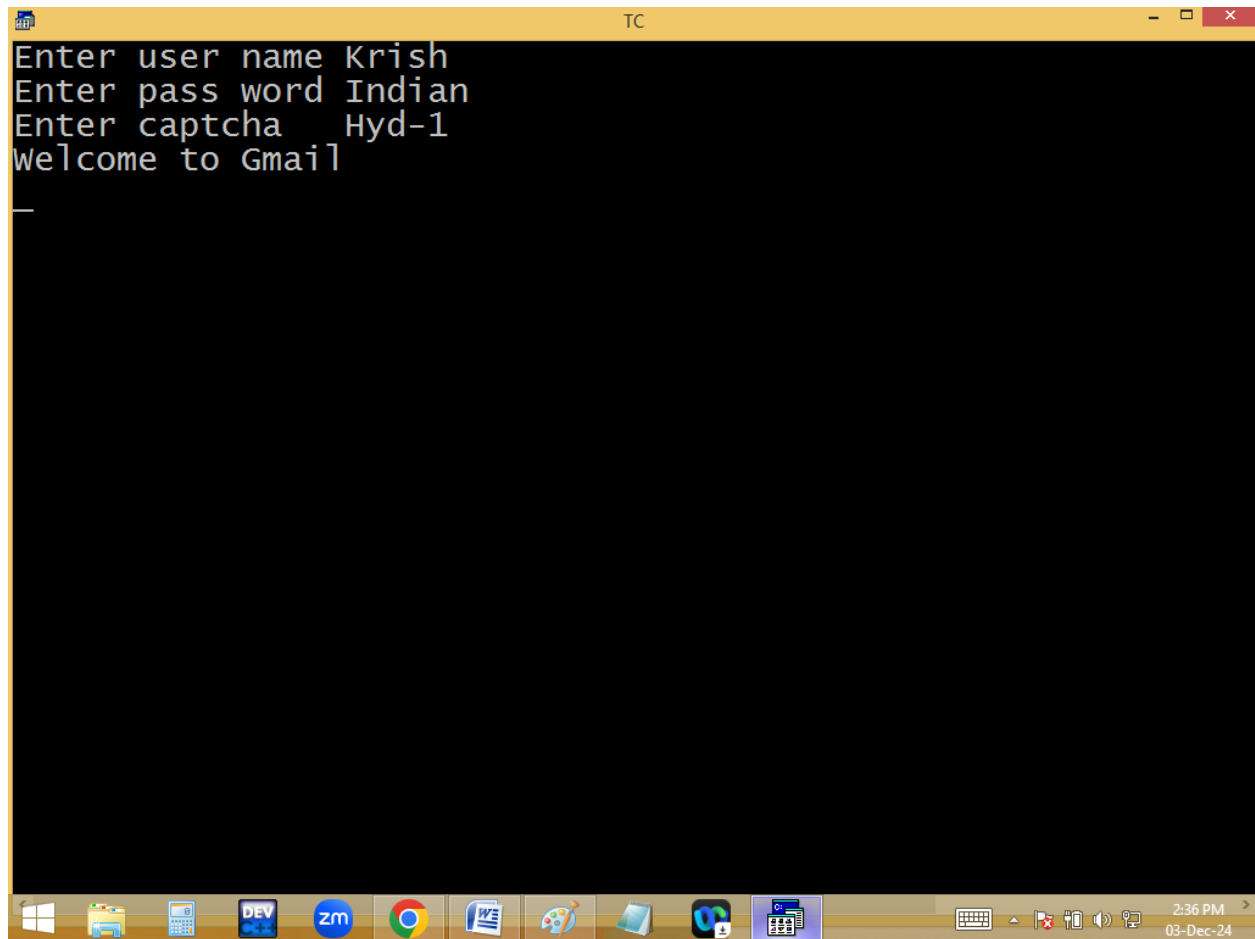
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

2:36 PM 03-Dec-24







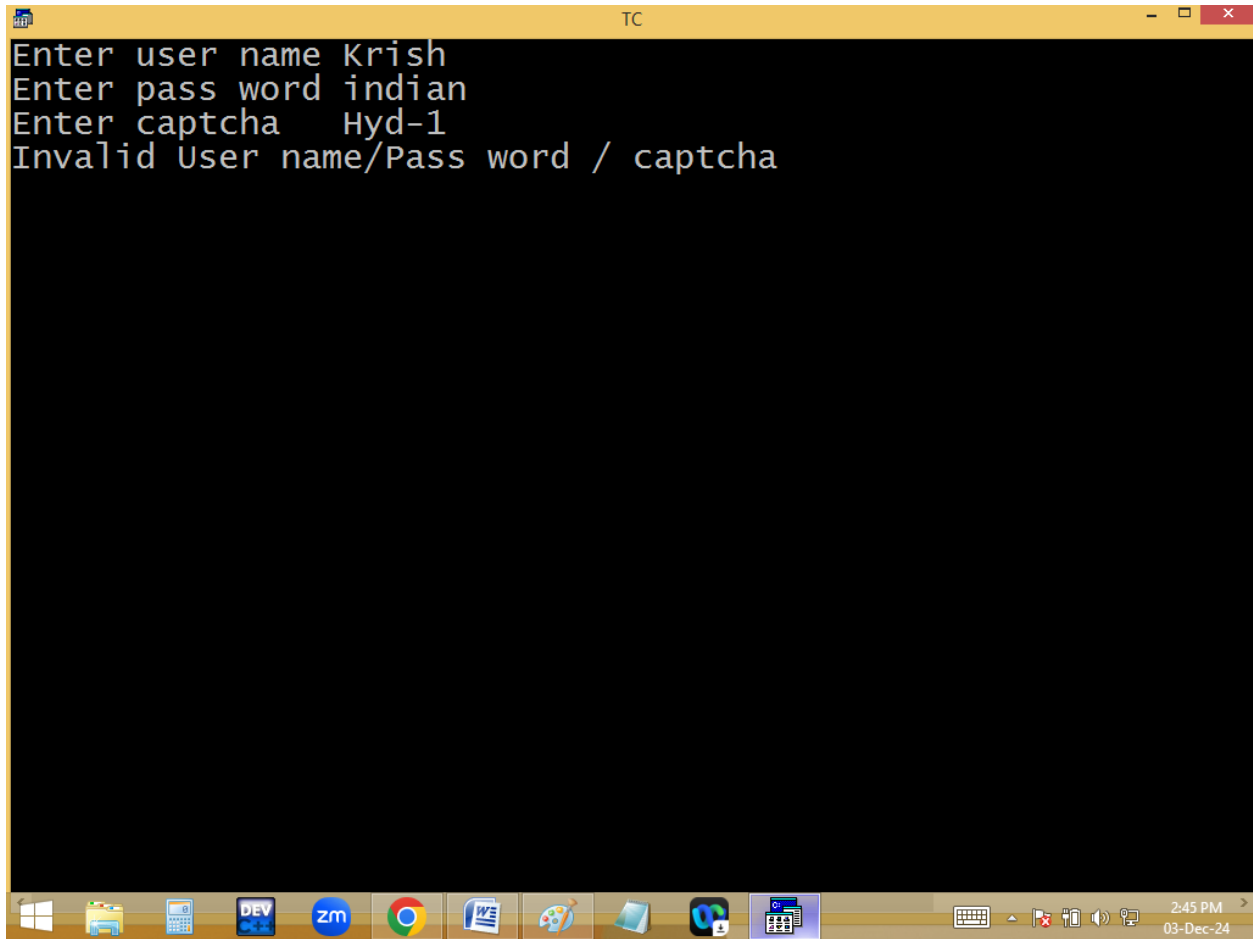


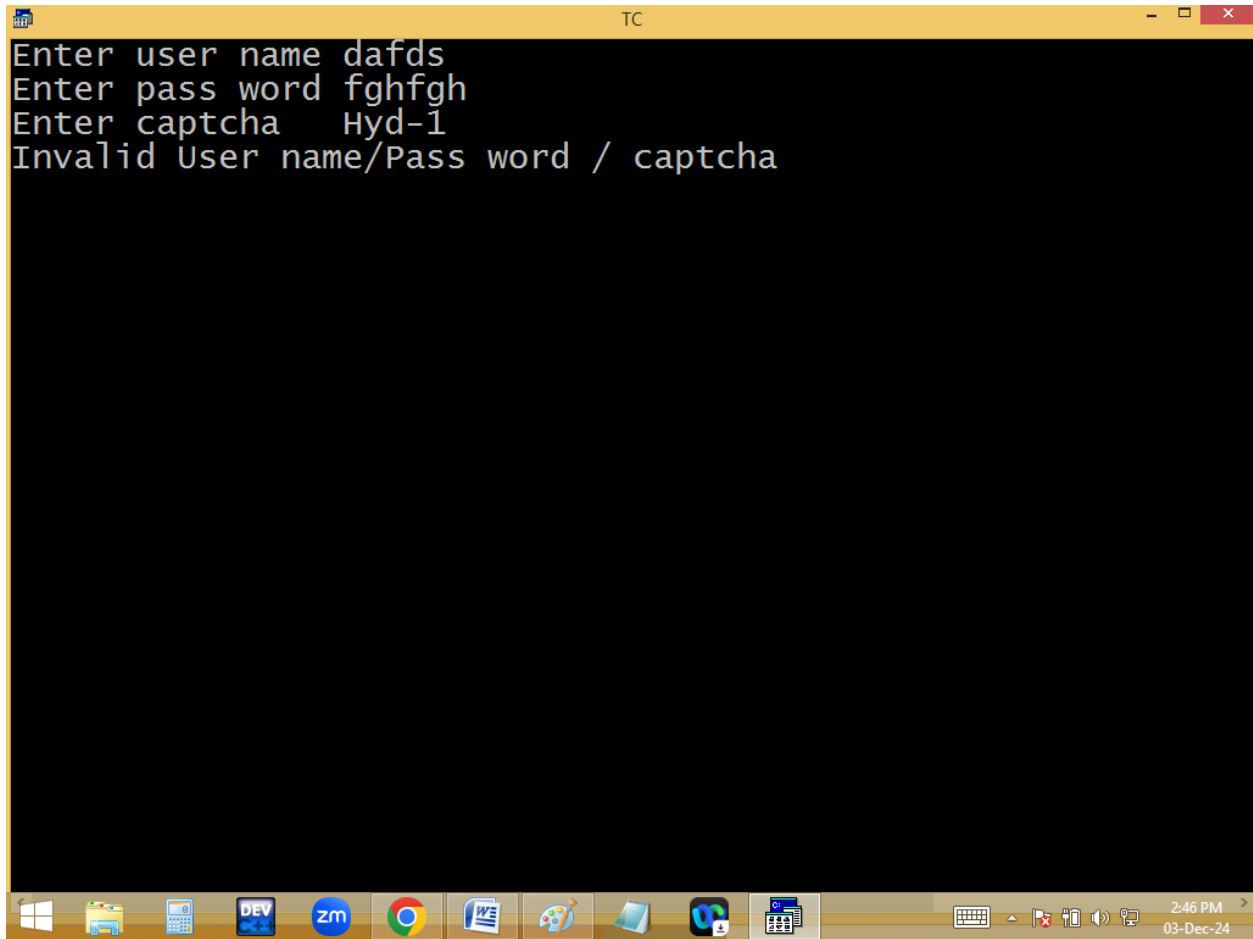
Net banking style login:

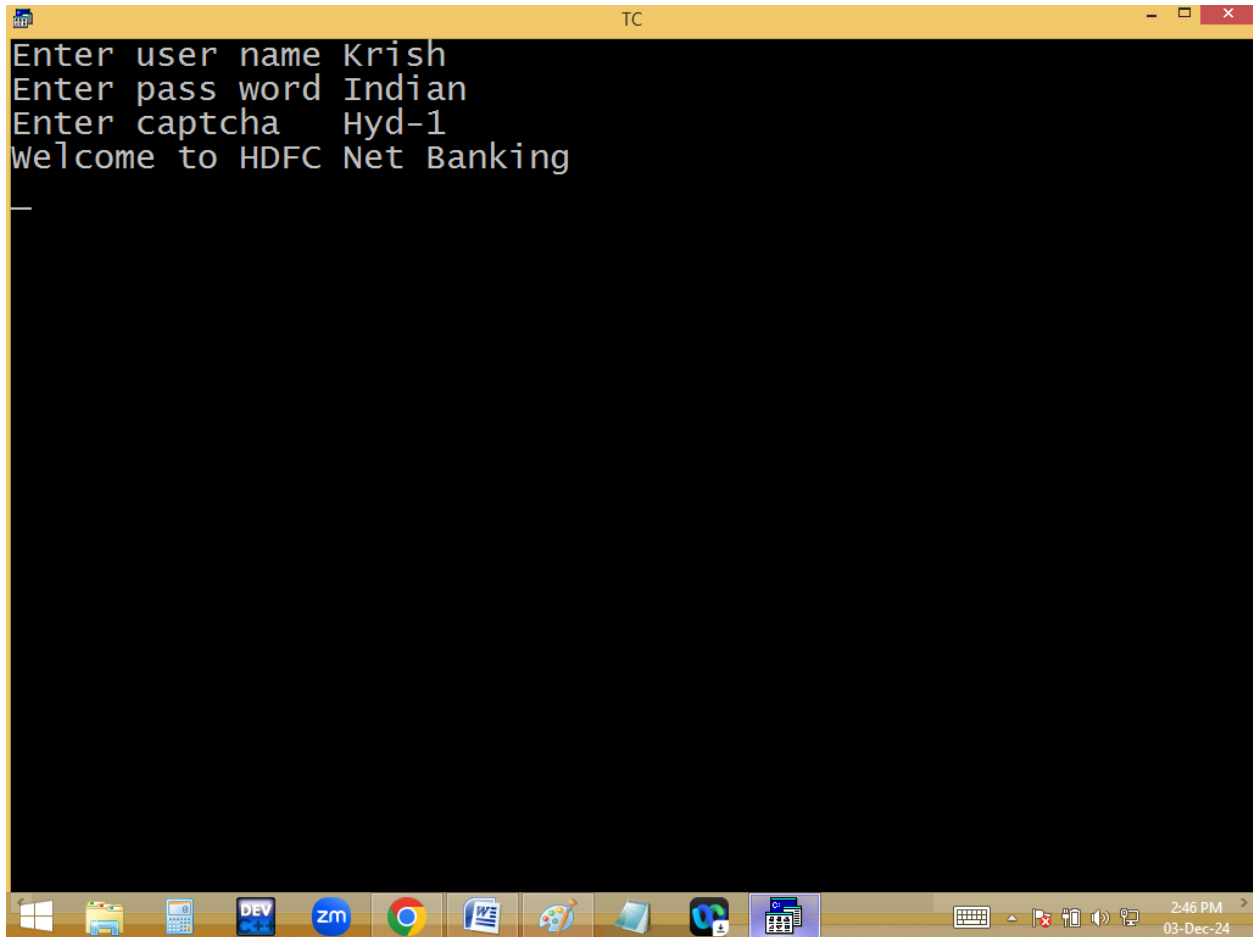
```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char user[20], pass[20], cap[20];
clrscr();
printf("Enter user name "); scanf("%s",user);
printf("Enter pass word "); scanf("%s",pass);
printf("Enter captcha "); scanf("%s",cap);
if(strcmp(user,"Krish")==0&&strcmp(pass,"Indian")==0&&
strcmp(cap,"Hyd-1")==0)puts("Welcome to HDFC Net Banking");
else puts("Invalid User name/Pass word / captcha");
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

2:45 PM 03-Dec-24







Read a stu id, name, 6 sub marks. Find total, avg and grade using below table.

Avg	Grade
≥ 75	Distinction
≥ 60	1 st class
≥ 50	2 nd class
≥ 35	3 rd class
< 35	Fail

```
#include<stdio.h>
```

```
#include<conio.h>
```

```

void main()
{
int id, tel,eng,hin,mat,sci,soc,tot;
char name[20];
float avg;
clrscr();
printf("Enter stu id "); scanf("%d",&id);
flushall();
printf("Enter stu name "); gets(name);
printf("Enter 6 sub marks ");
scanf("%d%d%d%d%d%d",&tel,&eng,&hin,&mat,&sci,&soc);
tot=tel+eng+hin+mat+sci+soc;
avg=tot/6.0;
puts("Id\tName\t\tTot\tAvg\tGrade");
puts("-----");
printf("%d\t%-17s%d\t%.2f\t",id,name,tot,avg);
if(tel>=35&&eng>=35&&hin>=35&&mat>=35&&sci>=35&&soc
>=35)
{

```

```
if(avg>=75)puts("Distinction");  
else if(avg>=60)puts("1st Class");  
else if(avg>=50)puts("2nd Class");  
else puts("3rd Class");  
  
}  
  
else puts("Failed");  
  
getch();  
  
}
```




```
TC
Enter stu id 1
Enter stu name Kishore naidu
Enter 6 sub marks 99 89 90 89 98 88
Id      Name      Tot      Avg      Grade
-----
1      Kishore naidu    553     92.17    Distinction
```

```
TC
Enter stu id 2
Enter stu name Kumari
Enter 6 sub marks 55 45 64 33 45 56
Id      Name      Tot      Avg      Grade
-----
2       Kumari    298     49.67    Failed
```

Ternary / Conditional operator(? :)

It is similar to if else / ladder if in working style.

It allows to complete if else / ladder if in a single statement.

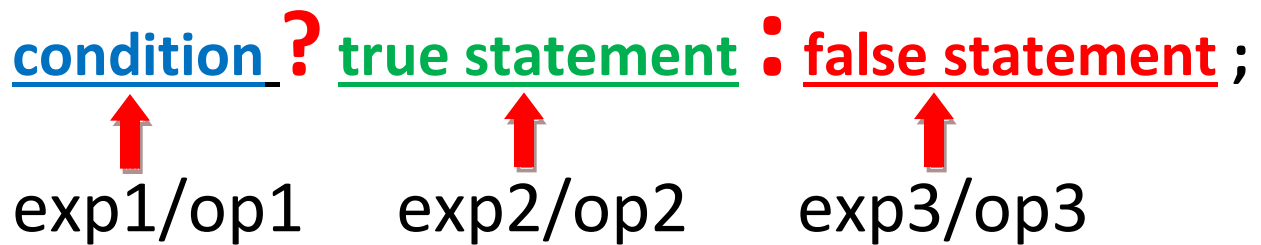
When we are working with if else/ladder if it is going to take more than one line of statements. Ternary operator is going to finish the same task in a single statement.

But the difference between if ...else and ternary operator is ternary operator supports only one statement at a time and if supports any number of statements.

It is having 3 expressions. Hence it is called ternary operator.

It is starting with a condition. Hence it is called conditional operator.

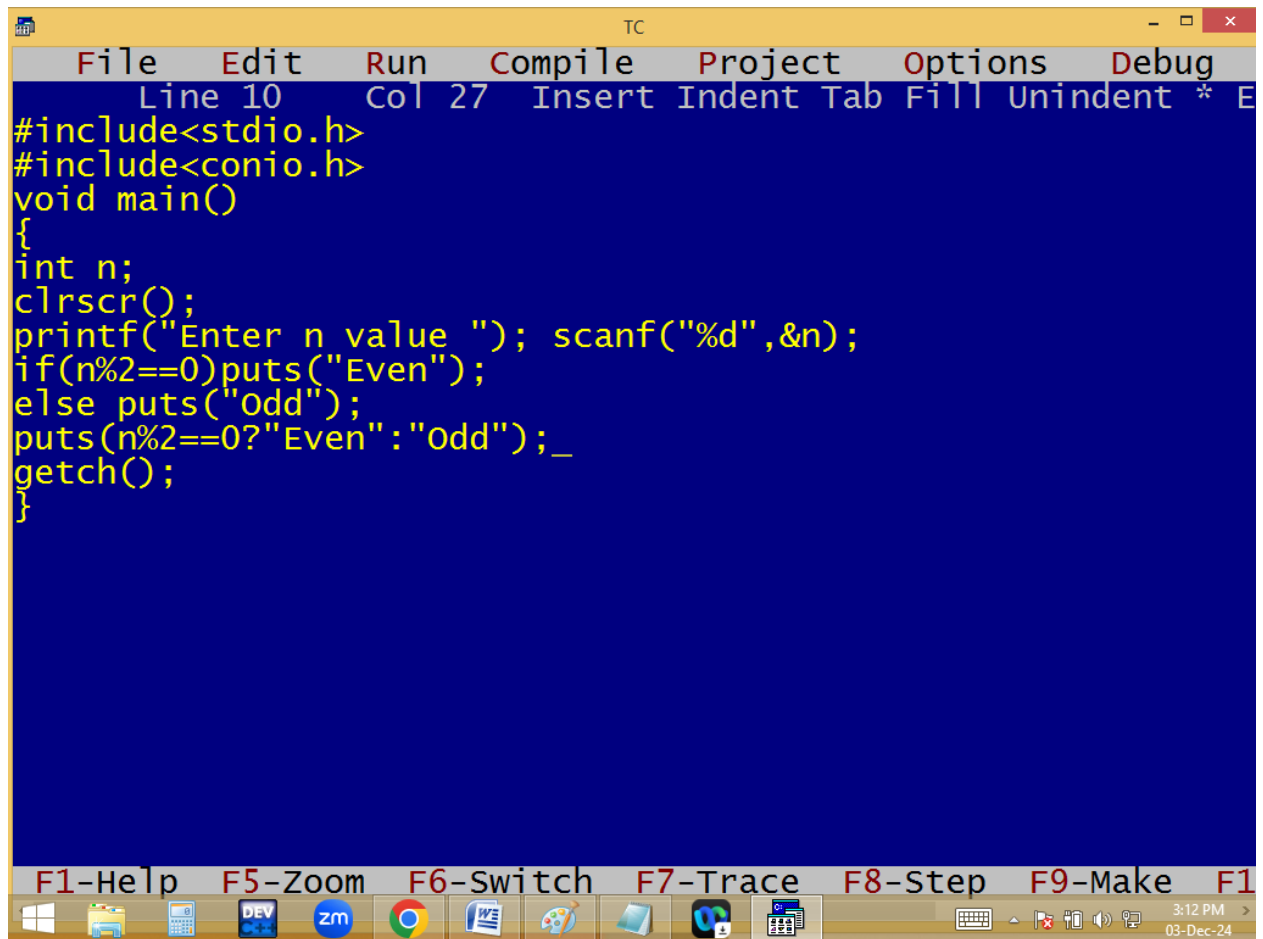
Syntax:

condition ? true statement : false statement ;

 exp1/op1 exp2/op2 exp3/op3

If condition true, statement after ? executed.

If condition false, statement after : is executed.

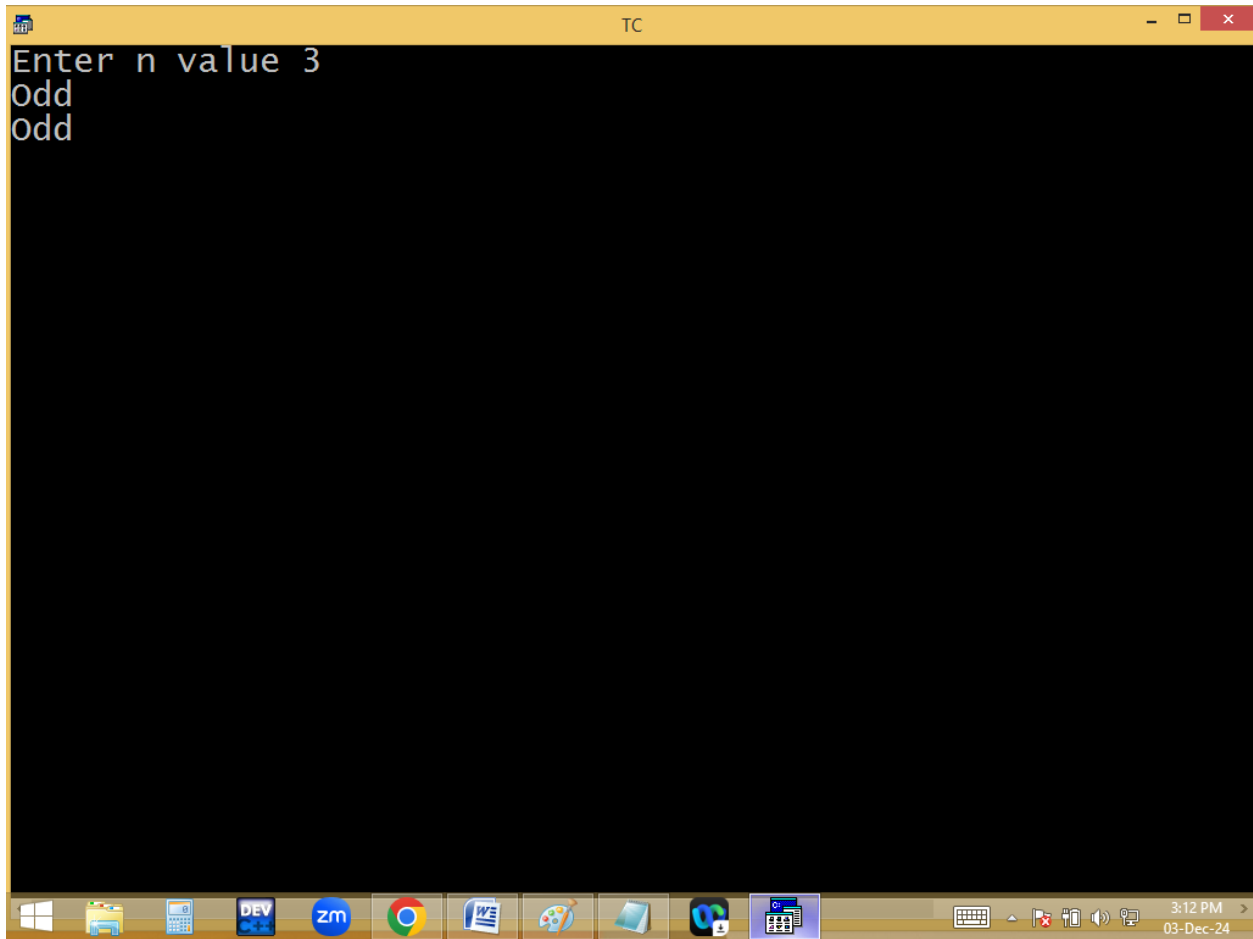
When compared with if else, conditional operator **performance is high**.



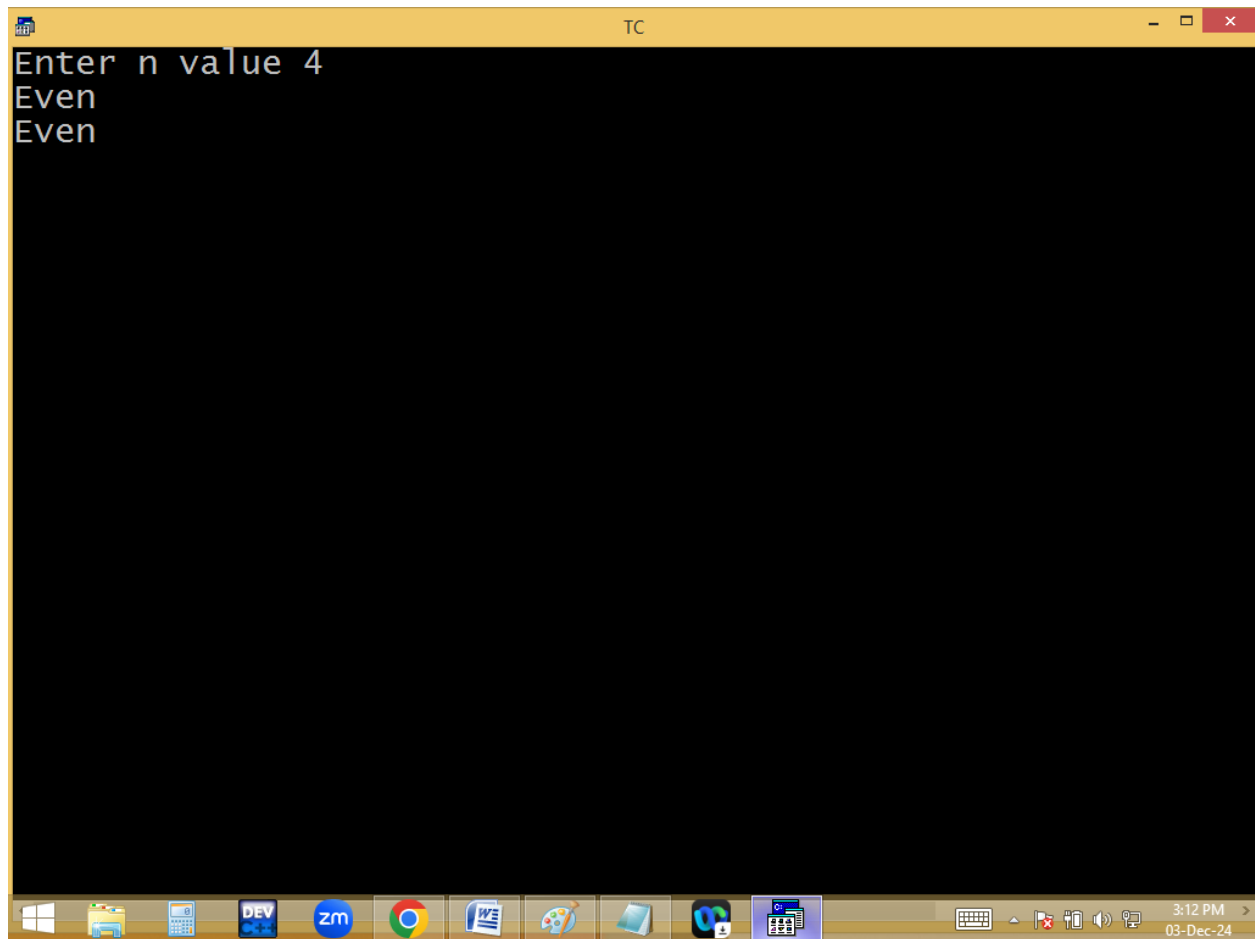
```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 27 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
if(n%2==0)puts("Even");
else puts("Odd");
puts(n%2==0?"Even":"Odd");_
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

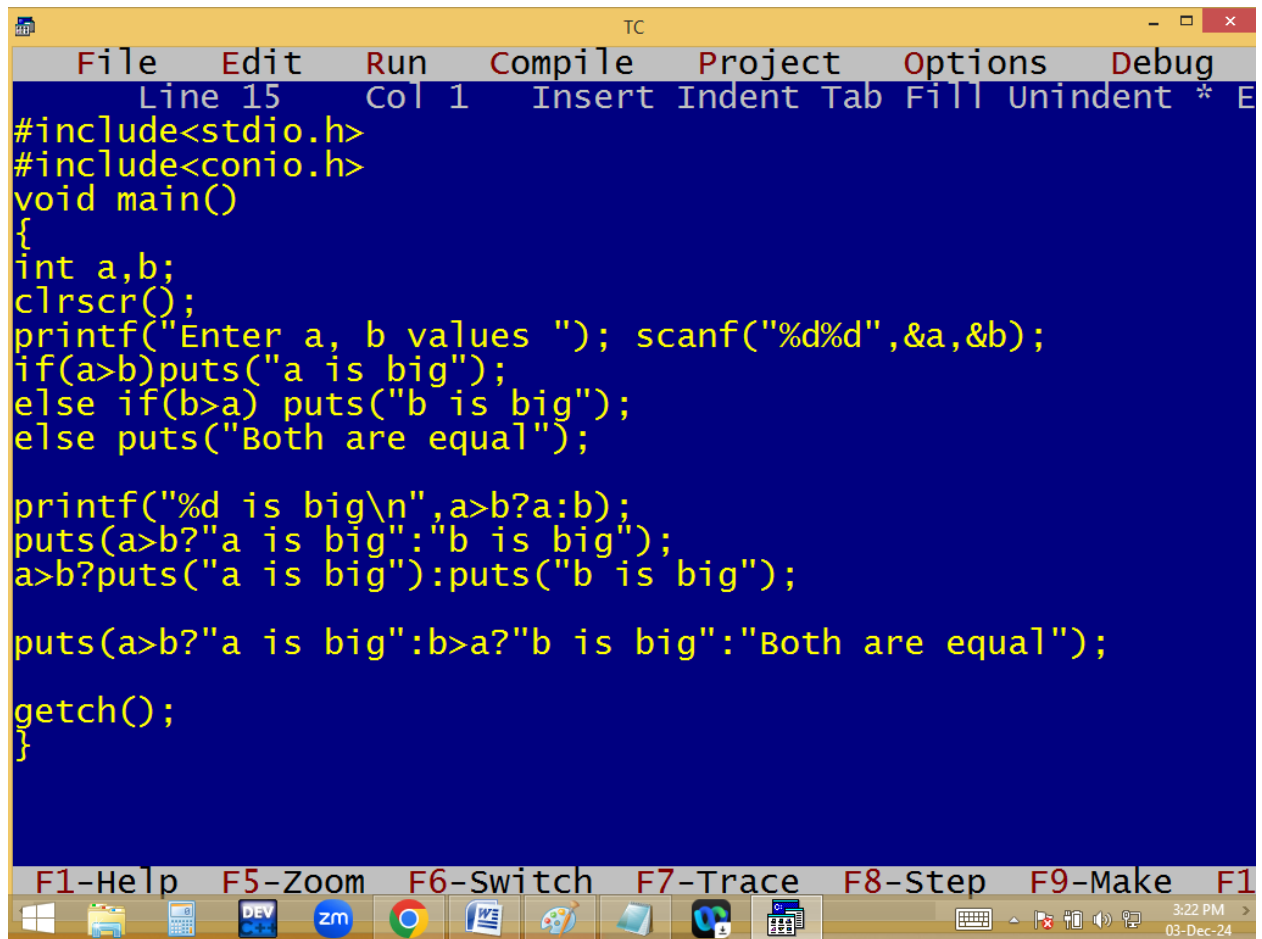
3:12 PM 03-Dec-24



```
TC
Enter n value 4
Even
Even
```



Finding max in 2 no's using ternary operator?



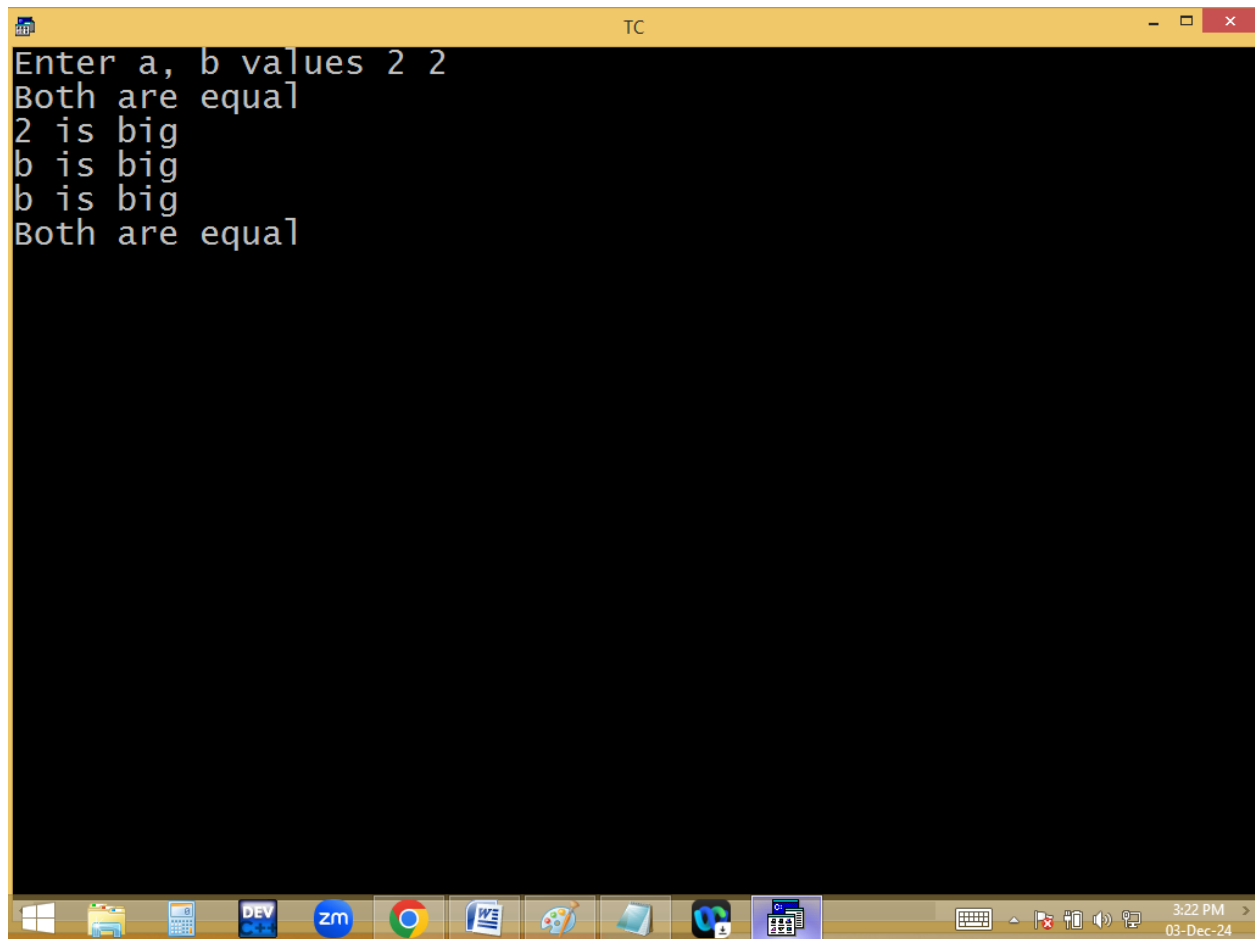
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a,&b);
if(a>b)puts("a is big");
else if(b>a) puts("b is big");
else puts("Both are equal");

printf("%d is big\n",a>b?a:b);
puts(a>b?"a is big":"b is big");
a>b?puts("a is big"):puts("b is big");

puts(a>b?"a is big":b>a?"b is big":"Both are equal");
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

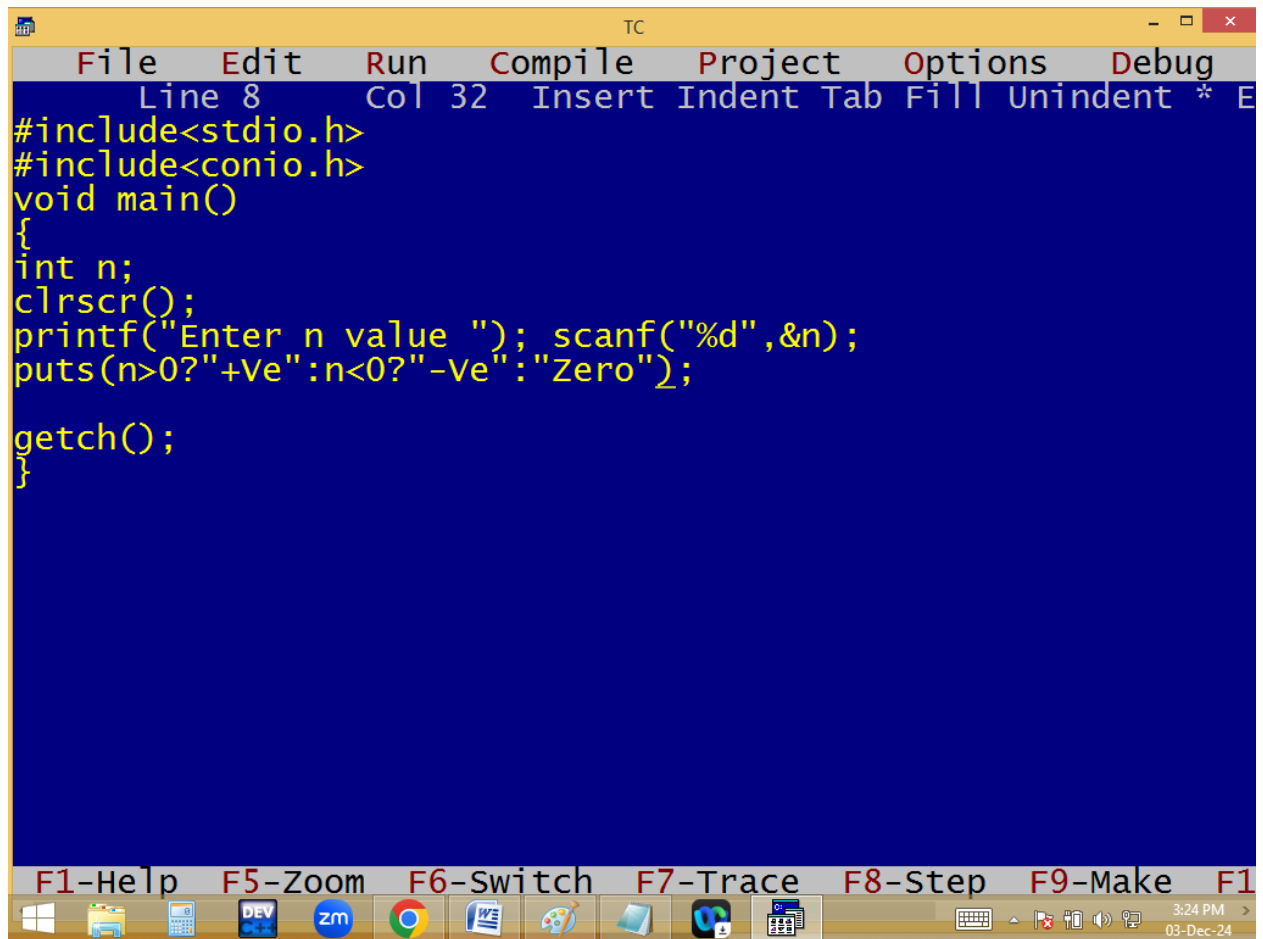
3:22 PM
03-Dec-24



```
Enter a, b values 2 2
Both are equal
2 is big
b is big
b is big
Both are equal
```

The screenshot shows a Turbo C++ (TC) compiler window. The title bar reads "TC". The main window area is black with white text. The text displayed is the output of a program that takes two input values, 2 and 2. The output consists of five lines: "Enter a, b values 2 2", "Both are equal", "2 is big", "b is big", and "b is big". The bottom of the window shows a Windows taskbar with various icons including the Start button, File Explorer, Calculator, DEV, zm, Chrome, Word, a game controller, a folder, a globe, and a calendar. The system clock in the bottom right corner shows "3:22 PM" and "03-Dec-24".

Finding +ve /-ve / 0 using ternary operator?

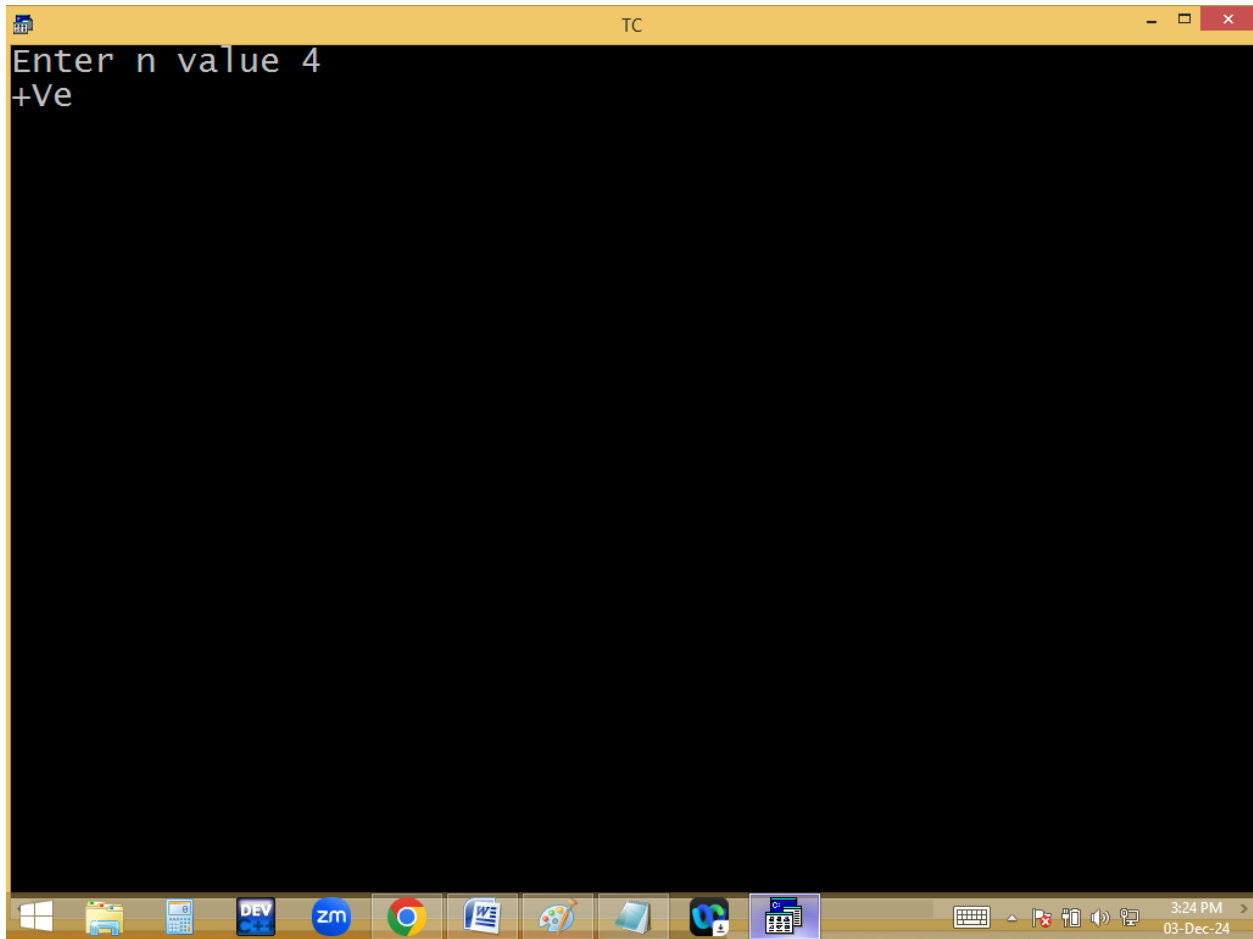


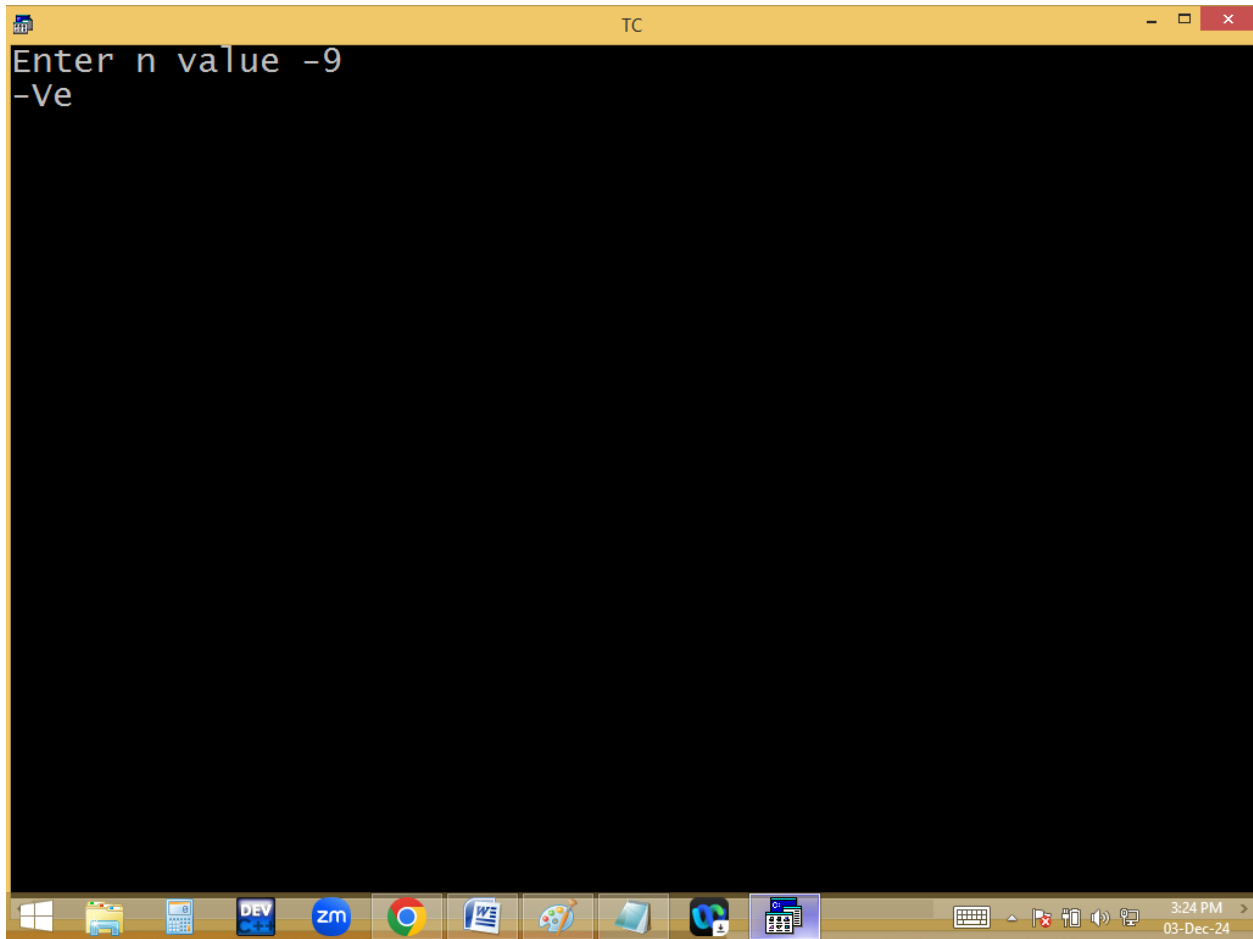
```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 32 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
puts(n>0?" +Ve":n<0?" -Ve":"Zero");

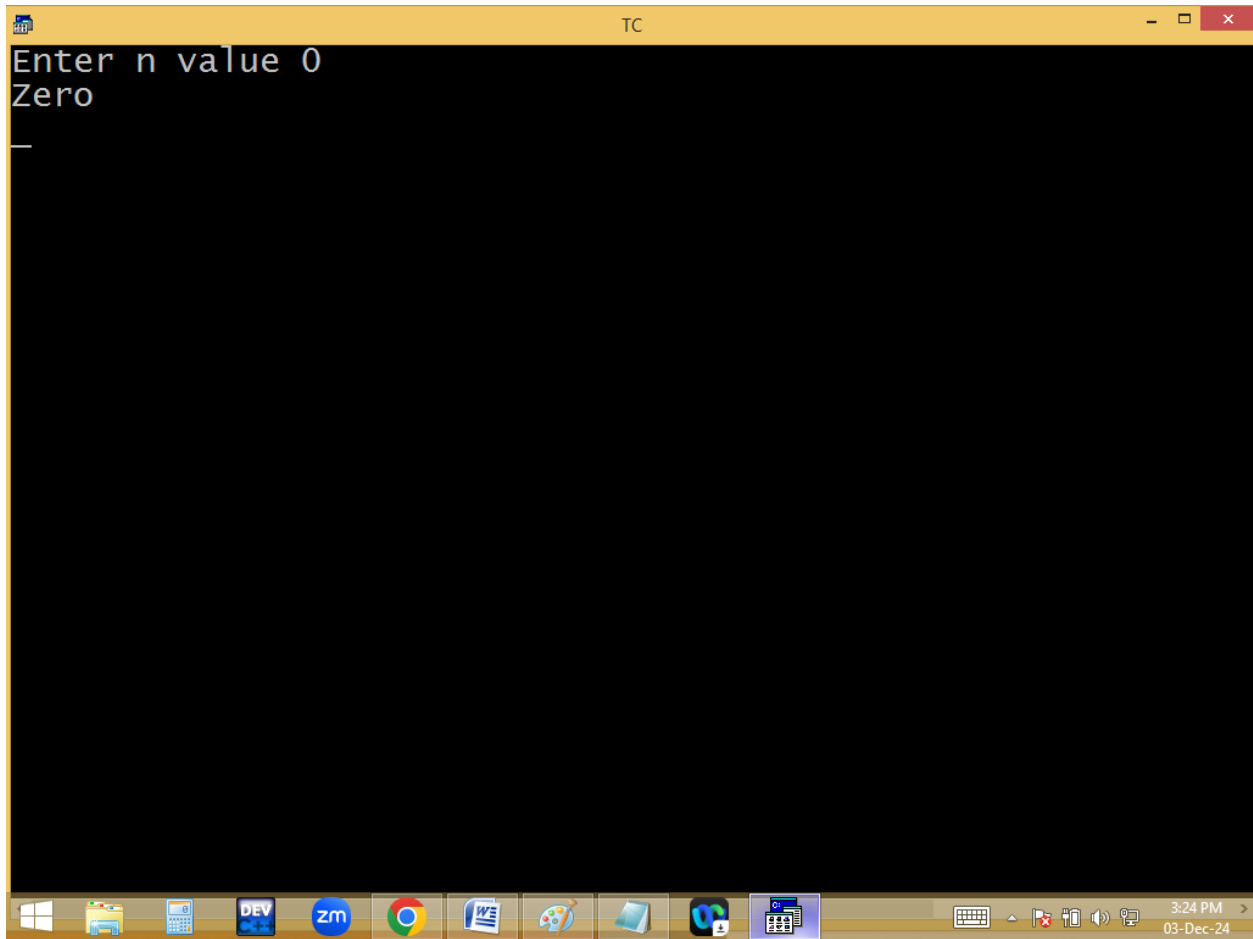
getch();
}
```

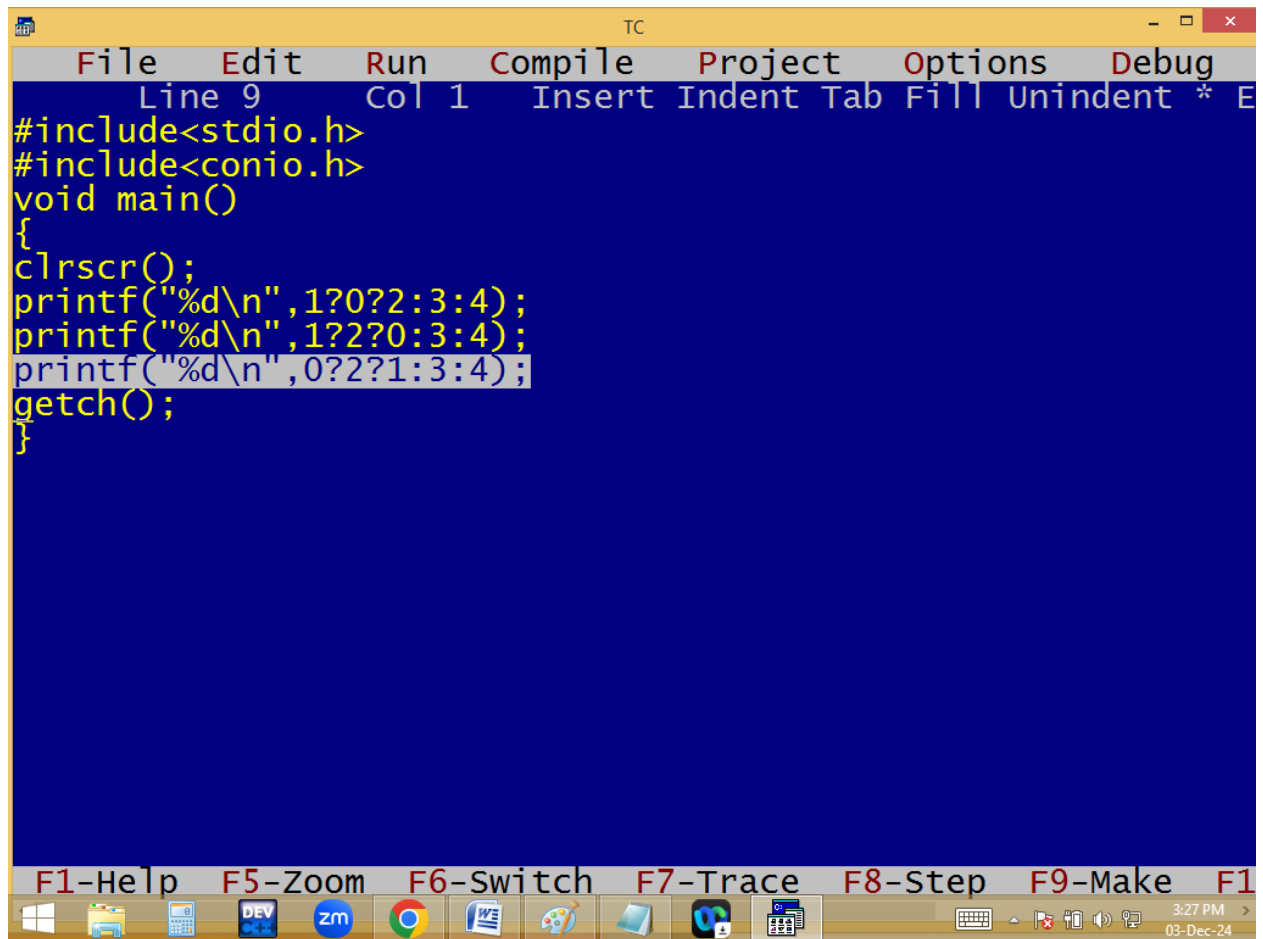
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

3:24 PM 03-Dec-24





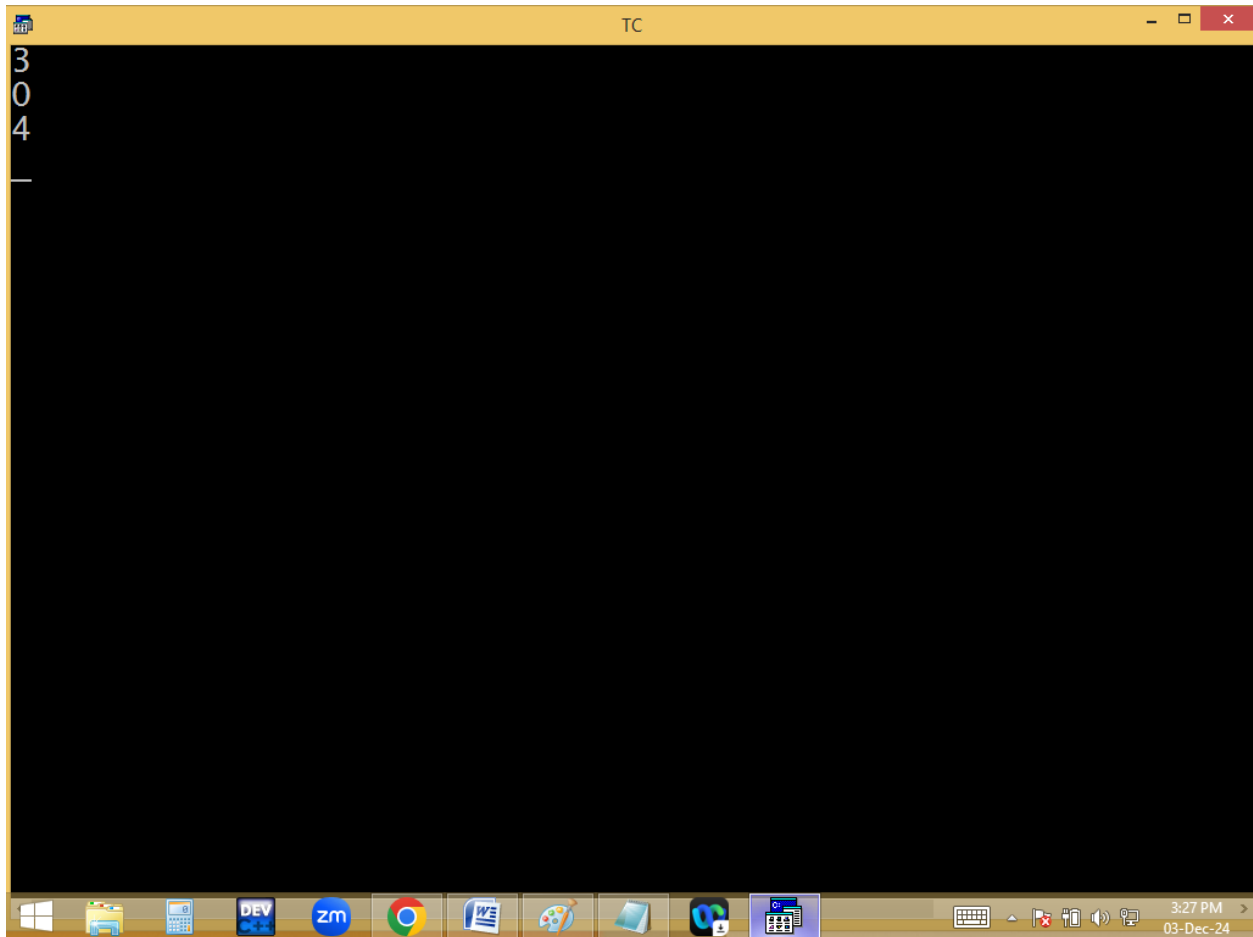




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 9 Col 1" and provides editing options: Insert, Indent, Tab, Fill, Unindent, and *. The code editor has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",1?0?2:3:4);
printf("%d\n",1?2?0:3:4);
printf("%d\n",0?2?1:3:4);
getch();
}
```

The third line of the printf statement is highlighted. The function key bar at the bottom includes F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar at the very bottom shows various application icons and the system clock indicating 3:27 PM on 03-Dec-24.



con ? true : false
1 ? 0 ? 2 : 3 : 4;
cond ? true part : false;

The diagram illustrates a nested ternary operator expression: `1 ? 0 ? 2 : 3 : 4;`. A red arrow points from the `con ? true : false` text above to the `0 ? 2 : 3` part of the expression. Another red arrow points from the `cond ? true part` text below to the `0 ? 2 : 3` part. The `0 ? 2 : 3` part is enclosed in a black rectangular box.

if(1)
{
 if(0) p(2);
 else p(3);
}
else p(4);

The diagram shows an if-else statement structure. A green arrow points from the `1` in `if( 1 )` to the `0` in `if( 0 )`, with a green 'T' (True) next to it. A red 'F' (False) is placed to the left of the `else` branch. A red checkmark is placed next to the `p(3);` statement, indicating it is the executed path.

con ? true : false
 1 ? 2 ? 0 : 3 : 4;
 cond ? true part : false;

```

if( 1 )
{
  if( 2 ) p(0);
  else   p(3);
}
else    p(4);
  
```

con ? true : false
 0 ? 2 ? 1 : 3 : 4;
 cond ? true part : false;

```

if( 0 )
{
  if( 2 ) p(1);
  else   p(3);
}
else    p(4);
  
```

goto label / jumping statement

It is used to transfer program execution from one place to another place [label].

In this process it is jumping from one area to another without any condition. Hence it is also called **unconditional** jumping statement.

Syntax:

```
.....;  
.....;  
goto label;  
.....;  
.....;  
label:  
.....;  
.....;
```

Here **goto** is a keyword.

Label is an identifier is used to identify the area[line].

Every label should be end with : (colon)

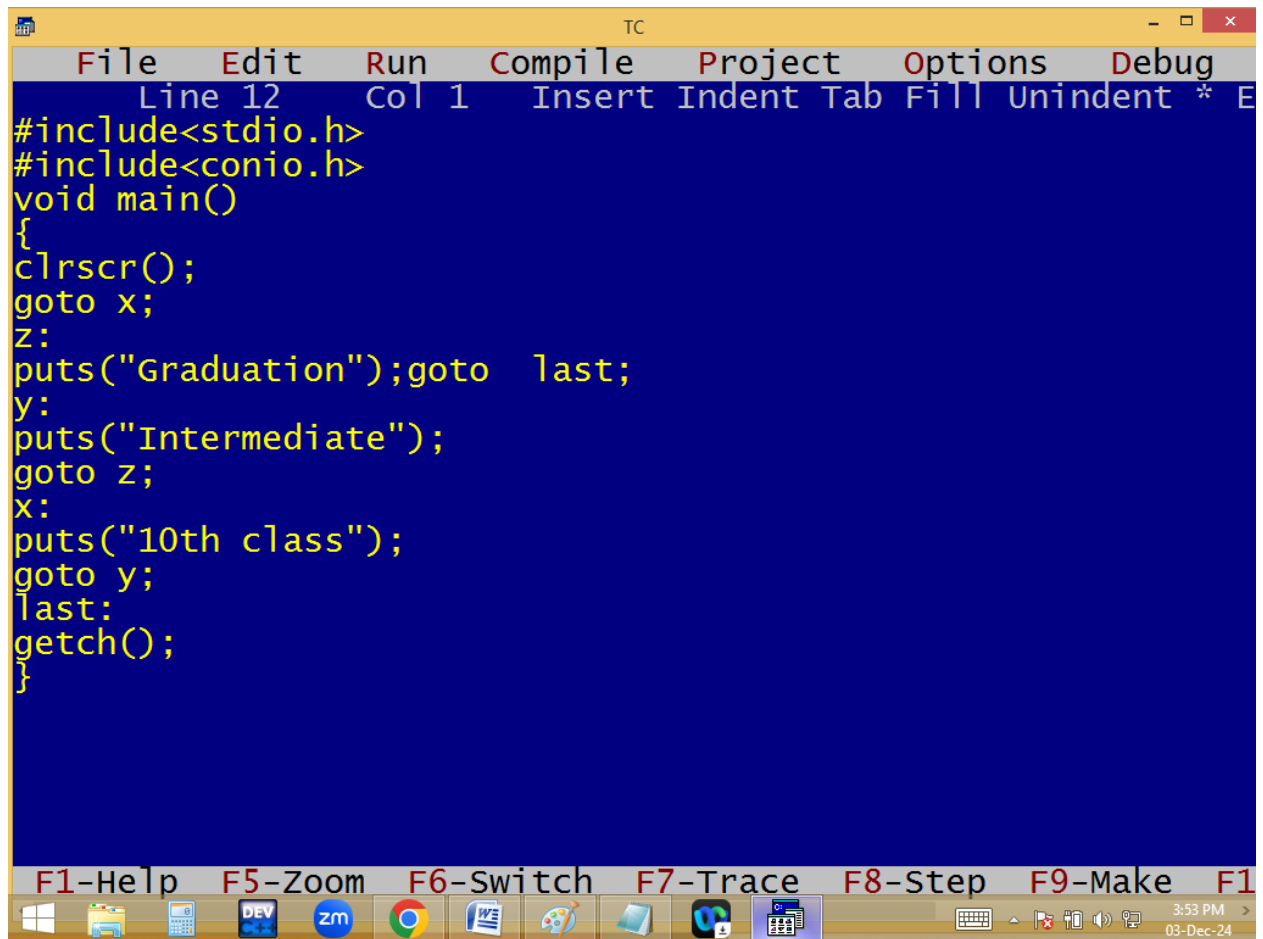
Keywords not allowed in labels i.e. label should be user defined.

Duplicate labels not allowed.

There is no space between go and to.

Label naming rules are similar to the identifier rules.

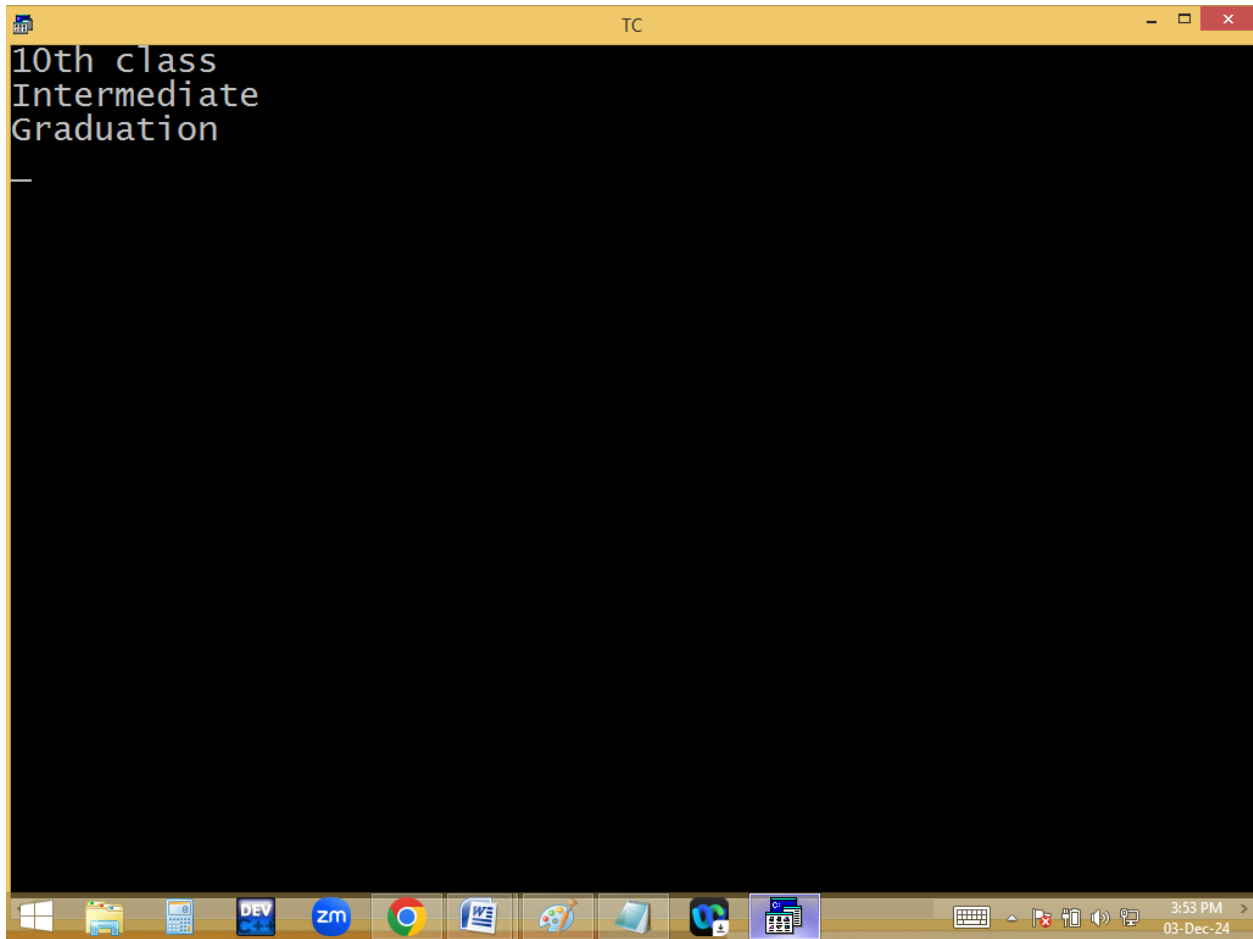
Note: goto label working style is similar to loops some times.

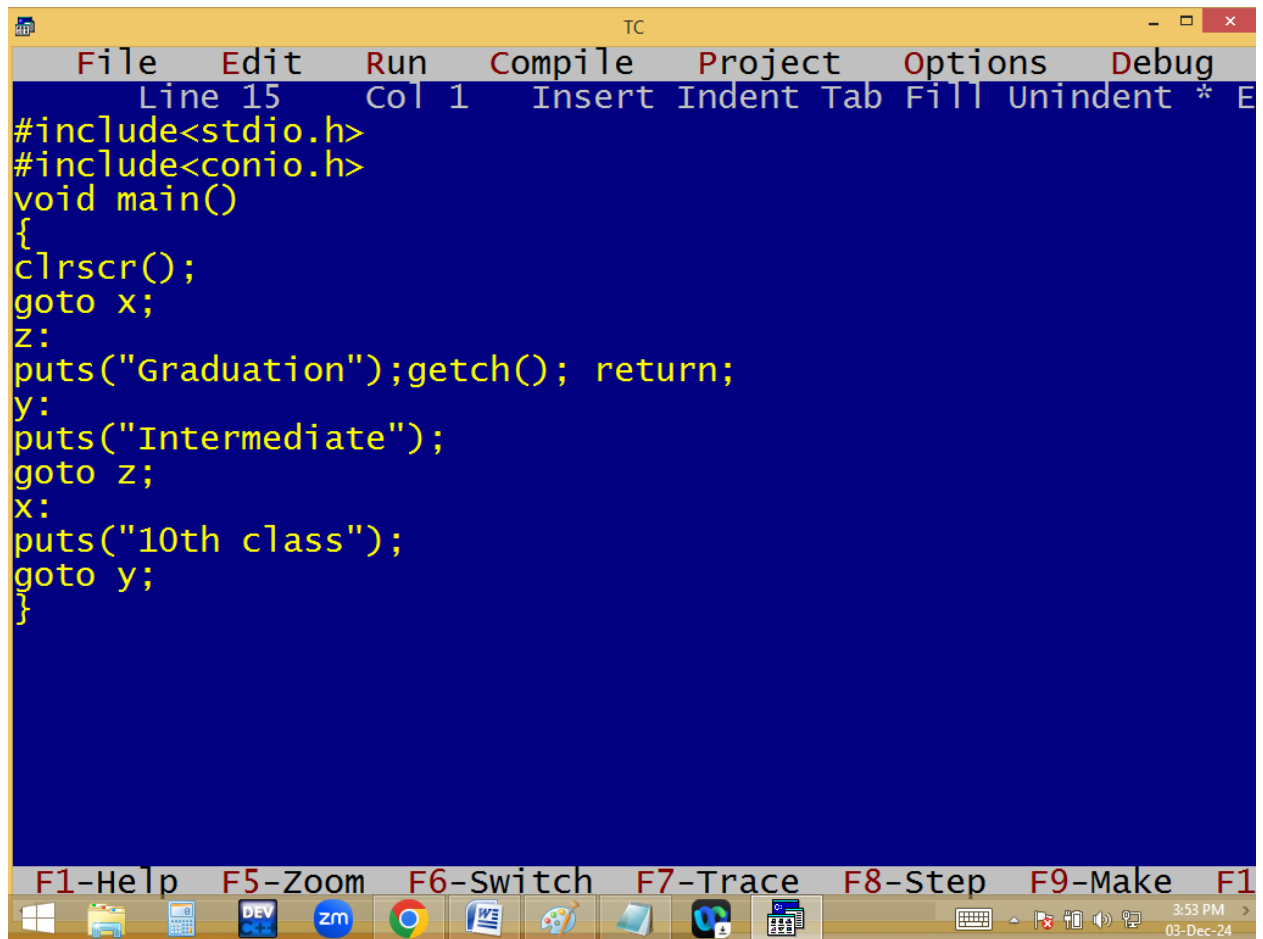


```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
goto x;
z:
puts("Graduation");goto last;
y:
puts("Intermediate");
goto z;
x:
puts("10th class");
goto y;
last:
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

3:53 PM 03-Dec-24

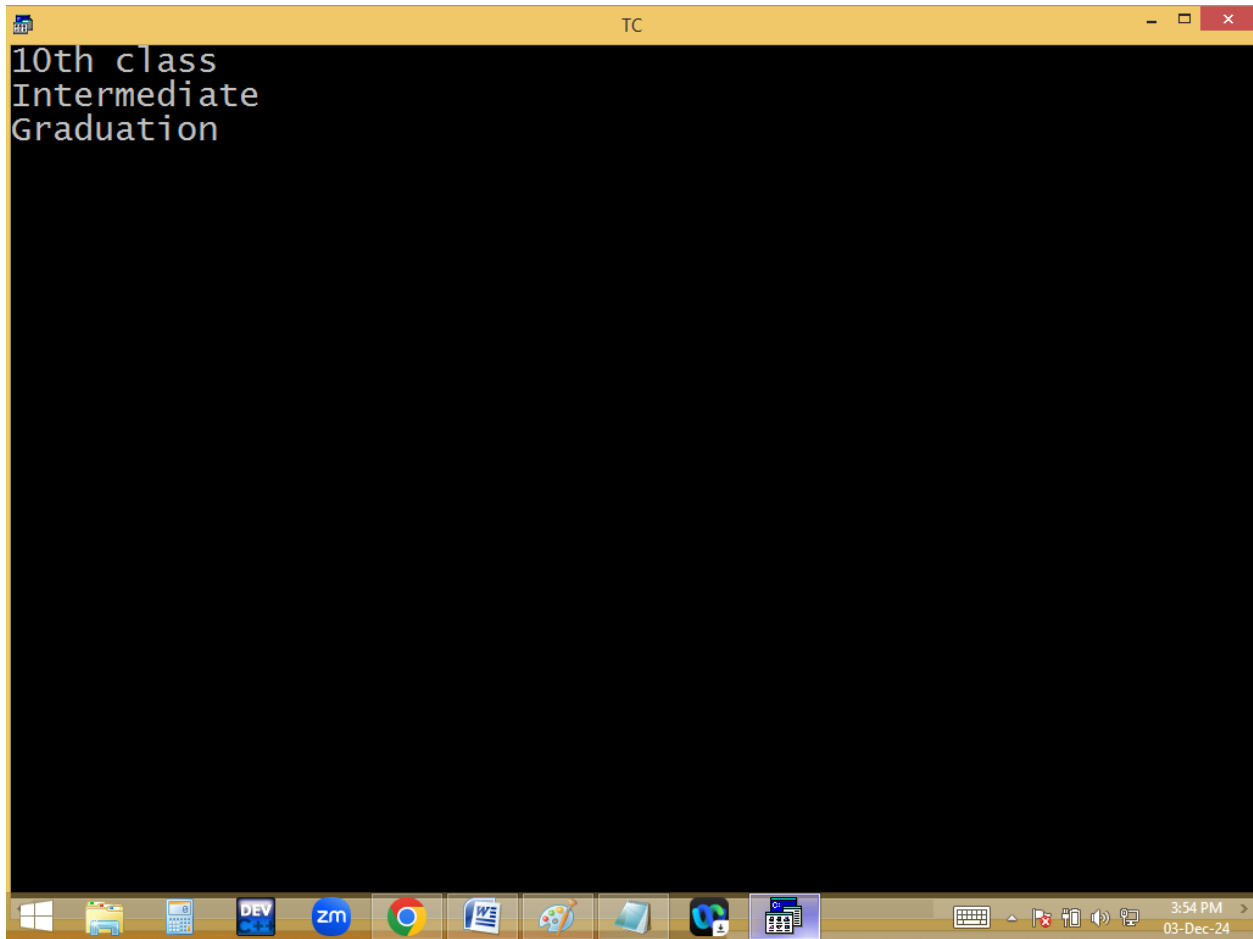


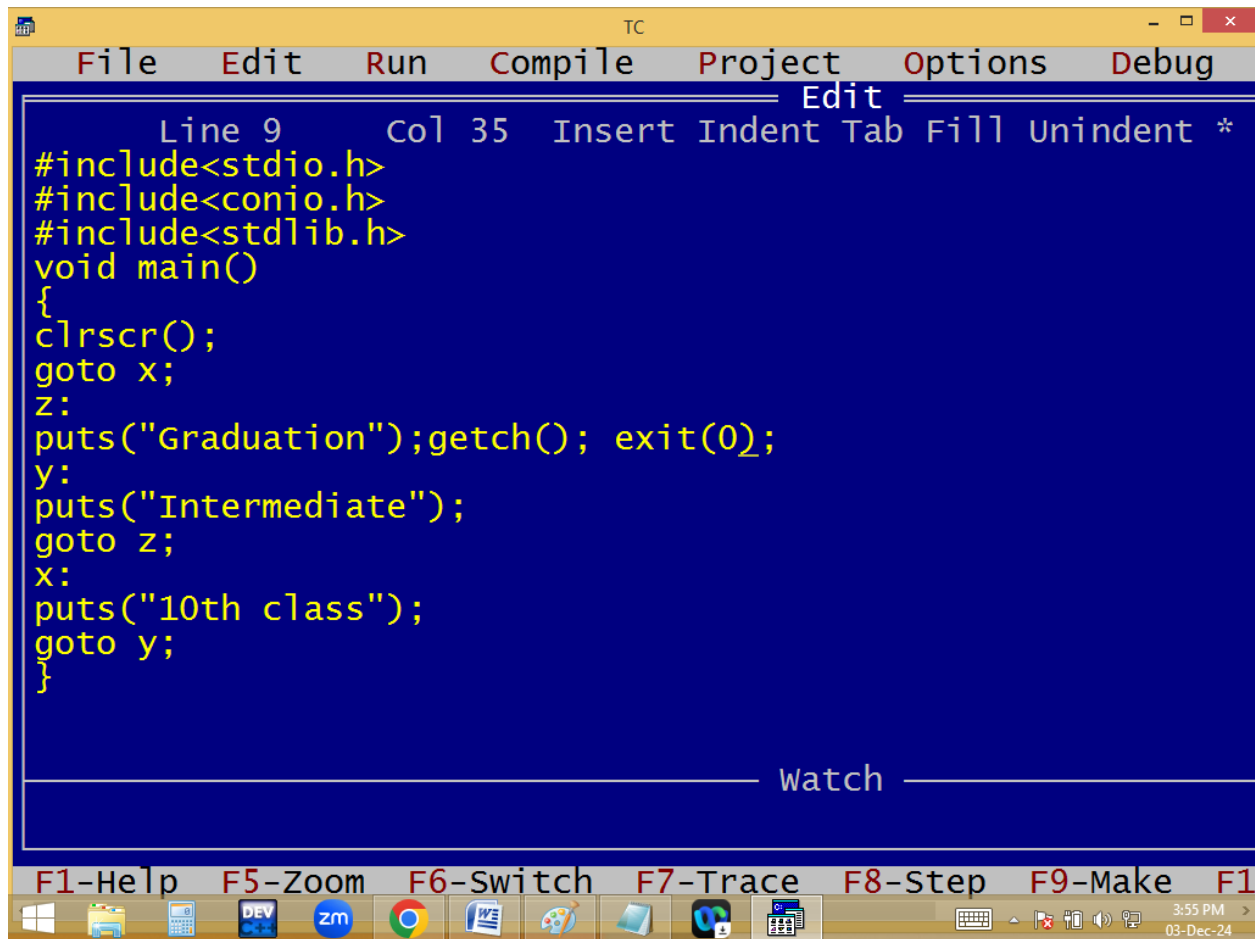


```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
goto x;
z:
puts("Graduation");getch(); return;
y:
puts("Intermediate");
goto z;
x:
puts("10th class");
goto y;
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F1

3:53 PM
03-Dec-24





The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The "Edit" menu is open, showing options: "Line 9", "Col 35", "Insert", "Indent", "Tab", "Fill", "Unindent", and "*". The main text area has a dark blue background with yellow text. It contains a C program that uses goto statements to print "10th class", "Intermediate", and "Graduation" in sequence. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
    clrscr();
    goto x;
z:
    puts("Graduation");getch(); exit(0);
y:
    puts("Intermediate");
    goto z;
x:
    puts("10th class");
    goto y;
}
```

Below the code editor is a "Watch" window, which is currently empty. At the bottom of the IDE, there is a toolbar with icons for various functions and a status bar showing the time "3:55 PM" and date "03-Dec-24".

Below the IDE window, there is a Windows taskbar with several icons: Windows Start button, File Explorer, Calculator, DEV C++, Zm, Google Chrome, Microsoft Word, Paint, Notepad, and a task view icon. The system tray on the right shows the time "3:55 PM" and date "03-Dec-24".

